The diagonal entries, or the variance of individual variables, are all ones in this case. Further, variables are ordered so that those that are close to each other are correlated; that is, their covariance takes a larger value. Variables that are far away from each other are, on the other hand, less correlated, and their covariances are close to zero. For example, any pair of consecutive variables in this Gaussian (the first and the second, the second and the third, etc.) have a covariance of roughly 0.87. That is, any two houses that are next to each other have a covariance of 0.87. If we consider the first and the twentieth variable, that is, the house at one end of the street and the house at the other end, their covariance is effectively zero.

This is very intuitive, as we expect houses that are close by to have similar prices, so once we know the price of a house, we gain more information about the prices of those that are around that area than about the prices of those that are far away.

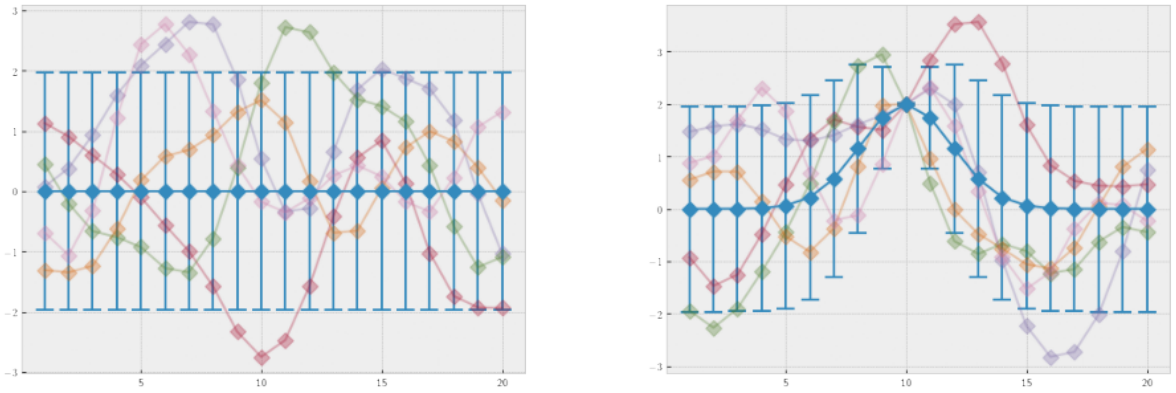


Figure 2.11 Error bars and samples drawn from a prior (left) and a posterior (right) Gaussian distribution, conditioned on the tenth variable having a value of 2. Uncertainty in variables that are close to the tenth reduces in the posterior, and their mean values are updated to be close to 2.

How does this play out in a parallel coordinates plot? Figure 2.11 shows our prior Gaussian on the left, and the posterior Gaussian conditioned on the tenth variable having a value of 2 on the right. Basically, we are simulating the event that we find out the price of the tenth house is 2 (whose exact unit is omitted).

- First, we once again see this phenomenon where in the posterior distribution, the error bars and samples are tied into a knot around the observation that we condition on.
- Also, due to the correlation structure imposed by the covariance matrix, variables that are close to the tenth have their mean values "dragged up" so that the mean vector now smoothly interpolates the point (10, 2). This means we have updated our belief in that the surrounding houses now have their prices increased to reflect the information we have learned.
- Finally, the uncertainty (denoted by the error bars) around this point (10, 2) decreases after the conditioning. This is a very good property to have, as intuitively, if we know the value of a variable, we should become more certain about the values of other variables that are correlated with the variable we know. That is, if we know the price of a house, we become more certain about the prices of nearby houses. This property is the basis of

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>